Mass Shootings in USA: Trends & Associated Factors

Department of Computer Science, Central Michigan University

Mahmudul Hasan Mamun, Jordan Leh, and Jordan Wood

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Supervisor: Dr. Michael Johnson

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1. Overview

Mass shooting has become a concerning issue in the United States of America since the frequency of mass shooting has increased in recent years. The country has witnessed 81 mass shootings in only the first two months of 2023 that resulted in 119 deaths and 329 injured cases (Source: Gun Violence Archive). So, it is important to explore the underlying reasons for this increasing trend of shooting and come up with recommendations to reduce it. This project aims to do an exploratory analysis of US mass shootings by combining data from different sources to answer some specific questions as follows:

- ✓ Identify significant factors associated with shootings.
- ✓ Analyze trends of mass shootings and other important socioeconomic factors of the mass shooting.
- ✓ Providing recommendations to reduce mass shootings.

US mass shootings- Kaggle (1966-2017)[1] and **US mass shootings- Kaggle (2018-2021)**[2], have US mass shootings data from 1966 to 2021. We created our **first dataset** by combining these two datasets that contain an assortment of variables including date, location, gender, age, race, fatalities, injured cases etc. Though Gun violence Archive has more recent years data (2014-2023), we did not merge them with this dataset as the two sources use different definitions of mass shootings to record data.

Our second dataset was created by aggregating yearly data on mass shootings from the first data and merging them with historical data of different important socio-economic factors. '**Gun Ownership in America: 1973 to 2021'**[3] online report contains historical data on household gun ownership in the USA. The '**Poverty rate over time (1959-2020)'**[4] was used from the website in the footnote. **Income inequality data (Gini Index): 1967-2021**[5] was collected from United States Census Bureau. **Violent Movie and TV series: Crime and Serial Killer (1975-2023)**[6,7] data was collected from IMDB.

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1 https://www.kaggle.com/datasets/zusmani/us-mass-shootings-last-50-years
2 https://www.kaggle.com/datasets/zusmani/us-mass-shootings-last-50-years
3 https://www.vpc.org/studies/ownership.pdf
4 https://www.self.inc/info/poverty-rates-in-each-state/?fbclid=IwAR2ViV6vGEN6YSOQA7aUo-_qbKPs5V6mS5aL6gqhcbfSYB9jkYIo3yHeHNg
5 https://www.census.gov/data/tables/time-series/demo/income-poverty/historical-income-inequality.html
6 https://www.imdb.com/search/title/?title_type=feature,tv_movie,tv_series&user_rating=7.0,10.0&keywords=Serial+Killer+
7 https://www.imdb.com/search/title/?title_type=feature,tv_movie,tv_series&user_rating=7.0,10.0&keywords=Crime

2. Dataset Description

After the collection and merging of different datasets, we finally created the following two datasets: Dataset 1- Mass Shooting Cases (1966-2021), and Dataset 2- Aggregated Mass shooting data (1979-2020).

Dataset 1- Mass Shooting Cases (1966-2021):
This dataset contains mass shooting cases that took place from 1966 to 2021. The table below describes all the columns, column type and description of the columns. The most important features that we are interested in are shooters' demographic issues (age, gender, race, mental health), no. of fatalities, injured and victims from mass shootings. This dataset can be utilized for the following purposes:

- ✓ Visualize historical mass shooting data on US maps to identify the riskiest areas.
- ✓ Identify the most significant factors associated with shootings.

Table 1: Instances (rows) and Features (columns) of mass shooting cases (1966-2021) dataset.

Instances (rows): Shooting incidences **No. of instances (rows): 451**				
No.	**Features (Columns)**	**Type**	**Description**	**No. missing instances**
1.	S#	int64	Case (shooting) number	0
2.	TITLE	object	Title	0
3.	LOCATION	object	Location (city, state) of shooting	45
4.	DATE	datetime64[ns]	Date of shooting	0
5.	SUMMARY	object	Summary description of the shooting	0
6.	FATALITIES	int64	No. of fatalities	0
7.	INJURED	int64	No. of injured	0
8.	TOTAL VICTIMS	int64	No. of total victims	0
9.	AGE	object	Age of shooter	144
10.	MENTAL HEALTH ISSUES	object	Did the shooter have any mental health issues?	0
11.	RACE	object	Race of shooter	0
12.	GENDER	object	Gender of shooter	0
13.	LATITUDE	object	Latitude	20
14.	LONGITUDE	object	Longitude	20
15.	YEAR	int64	Year	0

Dataset 2- Aggregated Mass shooting data (1979-2020)

This dataset contains yearly mass shooting aggregated data with other important indicators like poverty, income inequality, gun ownership etc. The table below describes all the columns, column type and description of the columns. This dataset can be utilized for the following purpose:

- ✓ Exploring and comparing time series trends of the no. of mass shooting with poverty rate, poverty gap, gun ownership, overall ratings of serial killers/crime movies & tv series etc.

Table 2: Instances (rows) and Features (columns) of yearly mass shooting aggregated data (1979-2020).

Instances (rows): Year **No. of instances (rows): 43**				
No.	**Features (Columns)**	**Type**	**Description**	**No. missing instances**
1.	YEAR	int64	Year	0
2.	# SHOOTING	int64	No. of total shooting	0
3.	# FATALITIES	int64	No. of total fatalities	0
4.	# INJURED	int64	No. of total injured	0
5.	# VICTIMS	int64	No. of total victims	0
6.	% TOTAL POVERTY	object	% of poverty	0
7.	GINI INDEX	float64	Gini index as income inequality	0
8.	CRIME MOVIE AVERAGE-RATING	float64	Overall ratings of serial killers/ crime movies & tv series	0
9.	% HOUSEHOLD GUN OWNERSHIP	float64	% of households own gun	20

3. Exploratory Data Analysis

3.1. Mass Shooting Victims

A total of 448 mass shooting incidences were recorded from 1966-2021 (by the mass shooting definition of the author[8]). On average, around 13 people were a victim of each shooting case. Among them, there were 58 percent fatalities and 42 percent injured cases. However, the mental sufferings of the survivors and the near and dear ones of the victims cannot be measured by quantitative data.

Table 3: Numerical measures of the number of fatalities, injured and total victims **by mass shooting incidences.**

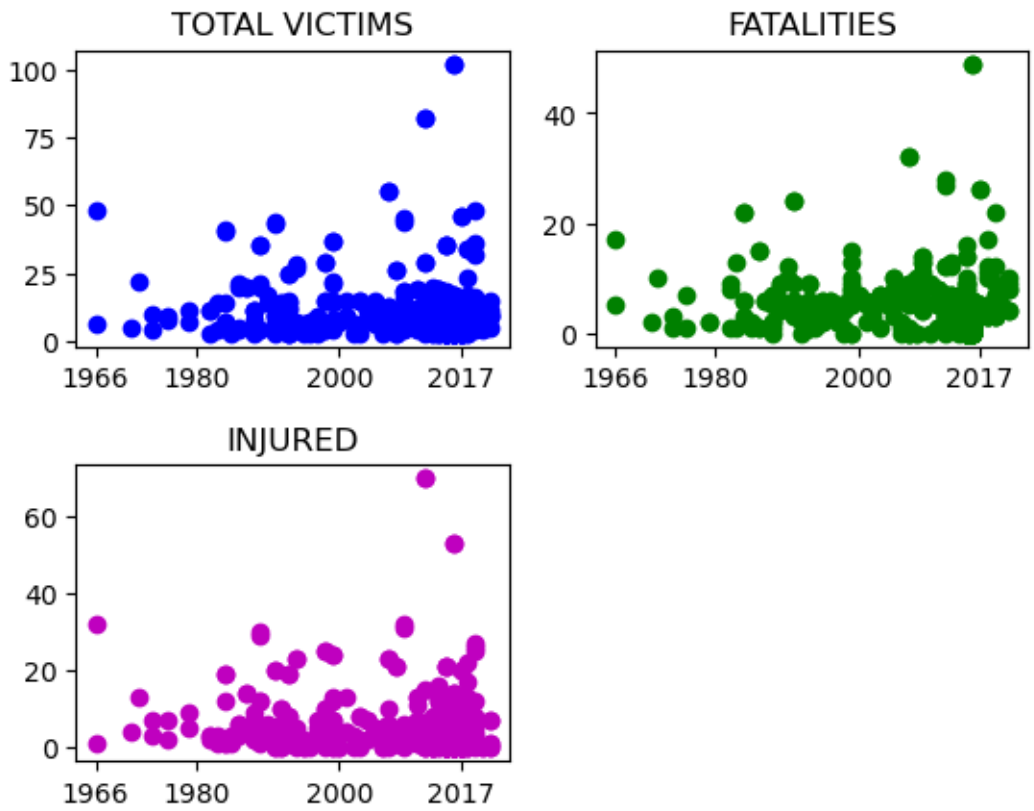
Numerical Measures	# of Fatalities	# of Injured	# of Total Victims
count	448.0	448.0	448.0
mean	5.4	7.7	12.9
std	6.6	36.4	40.8
min	0.0	0.0	3.0
25%	2.0	1.0	4.0
50%	4.0	3.0	6.0
75%	6.0	6.0	11.0
max	59.0	546.0	604.0

Scatter plots for each of the variables **(Figure 1)** are denser in recent years, which means the number of shootings has increased over time. We also observe an increased number of victims, fatalities and injuries in recent years (**Figure 2**). It is mentioned-worthy that the deadliest two shooting incidences (total victims: 604 and 585), which are discarded from the scatter diagram for better visualization, occurred in the year 2017.

Number of mass shootings as well as victims and fatalities caused by the shootings has increased over time. A significant increase is observed in recent years.

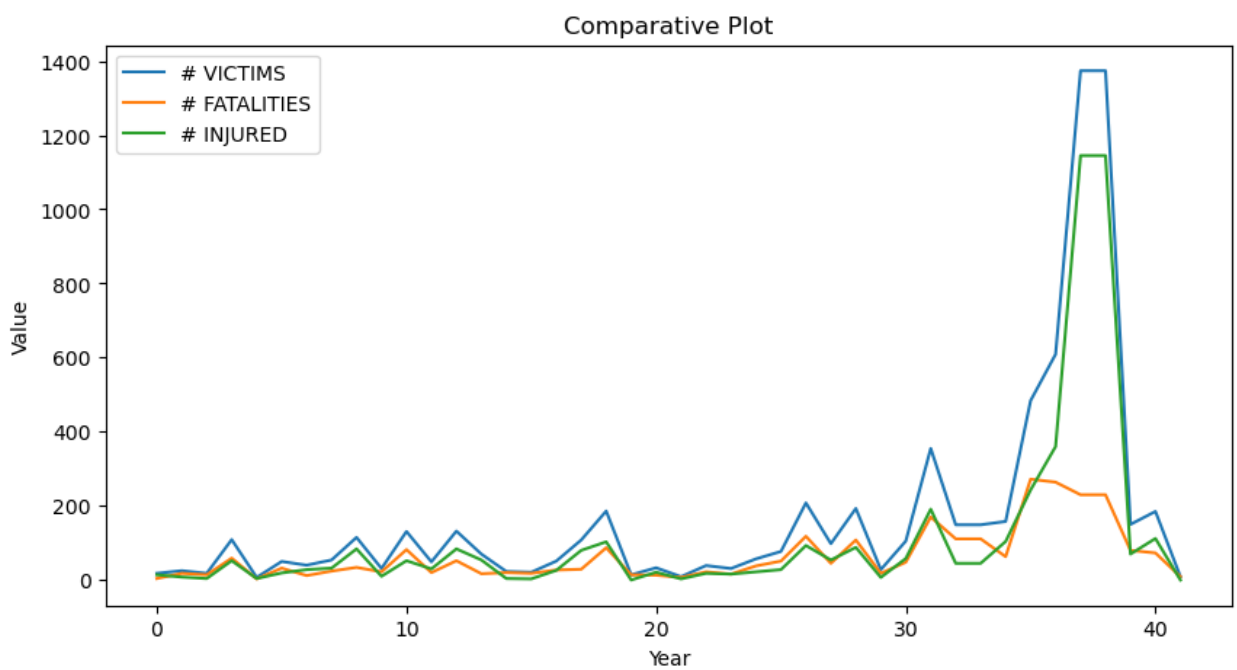
[8] Mass killing has no commonly agreed-upon definition. US Congress defines the mass shooting as a single shooting incident where three or more people are killed excluding the perpetrator. The gun violence archive, one of the largest and most dedicated sources to record mass shootings, includes all the shootings in which four or more people are shot or killed in a single incident excluding the perpetrator.

Figure 1: Scatter plot of no. of victims in each shooting case by year



Note: Largest two shooting cases (total victims: 604 and 585) were discarded from the scatter diagram for better visualization. Both incidents occurred in the year 2017.

Figure 2: Number of Victims, injured and fatalities **by year**



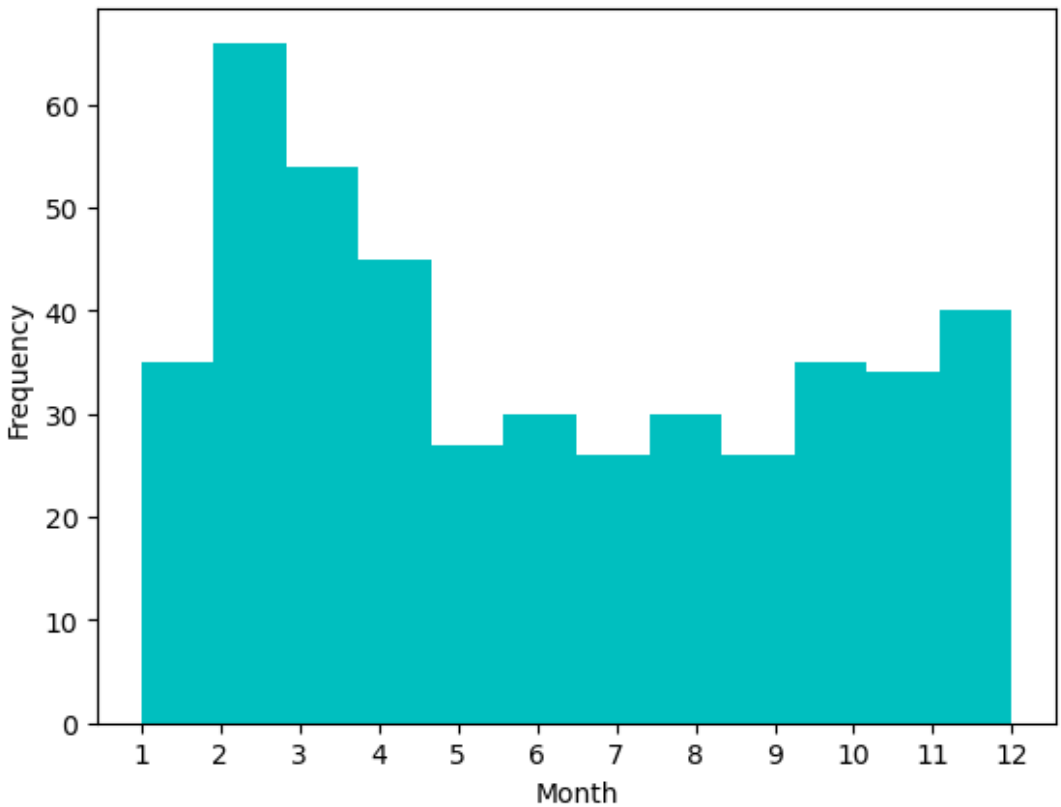
3.2. Factors Associated with Mass Shooting

Factor 1: Seasons of the Year

According to the histogram (Figure 3), more shootings occur in the beginning and end of the year. The highest number of shootings was observed in February, which is followed by March and April. In the middle months, we observe comparatively fewer shootings. This is obvious from the shape of the histogram.

Higher number of mass shootings were observed in February-April with the highest in February.

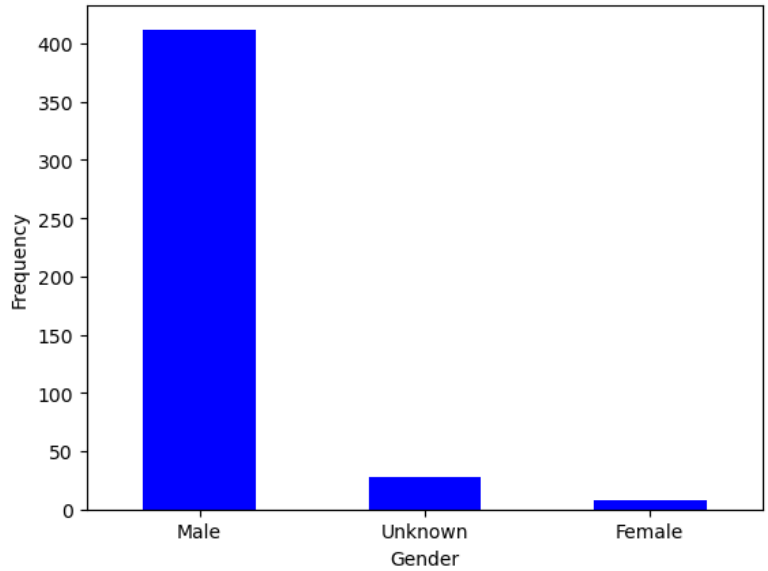
Figure 3: Mass shooting distribution by month (**January (1)- December (12)**)



Factor 2: Gender of the Shooter

According to **Figure 4**, most of the shooters were male (92%), while 8 percent were unidentified. Only 2 percent of the shooters were female.

Figure 4: Frequency of the shooter **by gender**

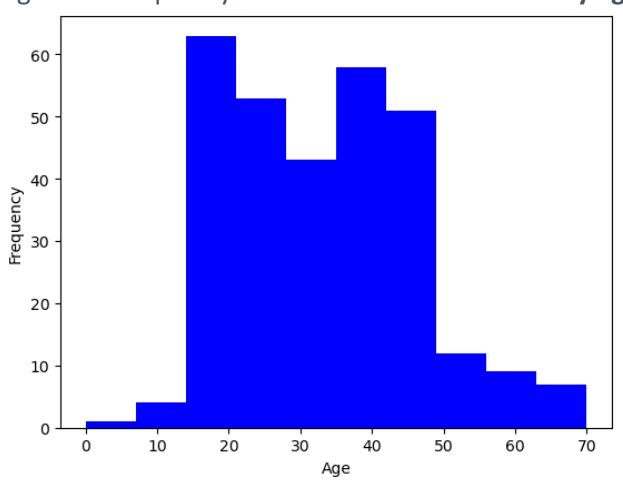


Factor 3: Age of the Shooter

The highest frequency of mass shooting is seen among Mid to late teenagers (15-20 years), at which this age of people are easily swayed by emotion. We also observed that a lion's share of the shooting cases occurs during young and middle adulthood. This is shown through a bell curved shape distribution of the histogram (**Figure 5)**.

Highest number of shooters were mid to late teenagers (15-20 years) with the majority in young and middle adulthood.

Figure 5: Frequency distribution of the shooter **by age**



Factor 4: Race of the Shooter

Half of the shooters were White American or European American, followed by Black American or African American (24%). The races of around 12 percent shooters are unknown. It is here to note that White American (including Hispanic and Latino) consist of 76% of the US population while Black or African Americans consists of 14%[9].

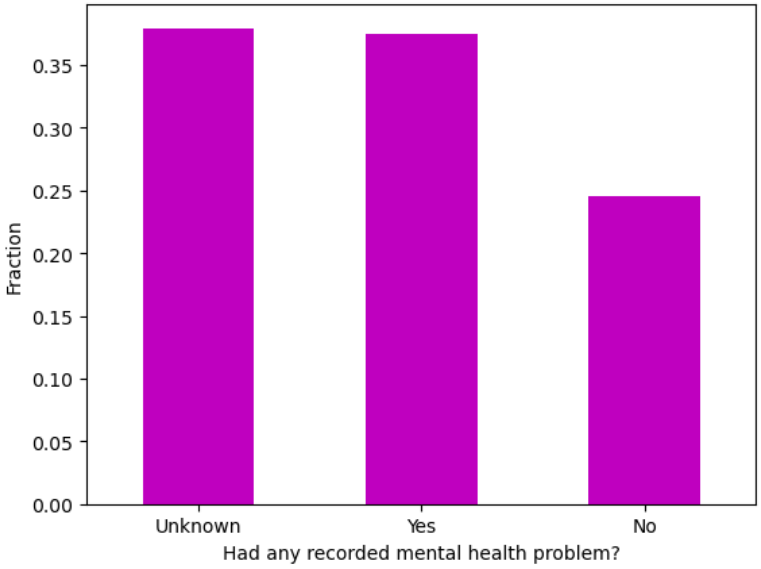
Table 4: Race distribution of the shooter (in %)

Race	**%**
White American or European American (including Hispanic and Latino)	50.0
Black American or African American	24.0
Unknown	12.0
Other races	14.0

Factor 5: Mental health problem

In the majority of cases, the mental health condition of the shooter was unknown (**Figure 6**). However, around one-third were suffering from mental illness, while one-fourth had no mental health issue.

Figure 6: Mental health problem of the shooter



[9] https://www.census.gov/quickfacts/fact/table/US/IPE120221

3.3. Time Series Trends of Mass Shooting and Important Socio-Economic Indicators

We observe that US mass shooting has increased over time with a significant recent increase (**Figure 7: Subplot- 1**). Historically, higher rates of mass shootings in US states are seen with more relaxed gun control laws[10]. So, it is widely believed that strict gun control will reduce mass shootings. However, interestingly, the percentage of households with gun ownership has reduced over time (32 percent in 1977 to 14 percent in 2021). We believe that stricter gun control may be a temporary solution, but we need to focus more on the rooted cause that might have triggered wide mass shooting. So, we analyzed some important factors that we believe can be the most significant underlying causes (obviously not limited to these only).

Gun control may reduce gun violence temporarily, but we believe that's not the permanent solution and we need to dig deeper.

It is evident from data that the US income inequality has increased over time (higher Gini Index indicates greater inequality: 0.351 from 1968 to 0.474 in 2021) (**Figure 7: Subplot- 2**). Recent studies indicate that income inequality triggers pessimism and anger among the US population[11][12][13].

If we look at the reported causes of the mass shooting cases (**Table 5**), we observe that the top causes include **anger, frustration and revenge**. This is very concerning! Also, terrorism is the second largest cause (19%) after mental health problems (21%) which might also be the indirect effect of inequality, anger and frustration. At this juncture, we hypothesize that income inequality is playing a vital role in increased mass shootings.

Income inequality causes anger and frustration, and may in turn trigger gun violence, which is supported by the reported causes of mass shootings from our data.

We further observe there are regular releases of good-rating violent movies in recent times, which may be a proxy indicator that more people like violent movies in recent days than before. While there is no consensus on the matter that violent movie increases crime, but almost half of the

Violent movies that involve attractive and hero-type characters may trigger children to emulate characters and contribute to violent behaviors like mass shooting.

[10] https://www.bmj.com/company/newsroom/higher-rates-of-mass-shootings-in-us-states-with-more-relaxed-gun-control-laws/
[11] https://www.capitalgazette.com/opinion/columns/ac-ce-column-perry-weed-20200823-tvcctkpnrzbihlztoe6q7cffce-story.html
[12] https://www.latimes.com/opinion/op-ed/la-oe-sapolsky-air-rage-and-inequality-20160710-snap-story.html
[13] https://www.americanprogress.org/article/income-inequality-in-the-united-states-fuels-pessimism-and-threatens-social-cohesion/

violent scenes on TV involve attractive, hero-type characters worthy of emulation[14]. So, children watching those movies may want to emulate the characters.

Figure 7: Time Series Trends of Mass Shooting and Important Socio-Economic Indicators

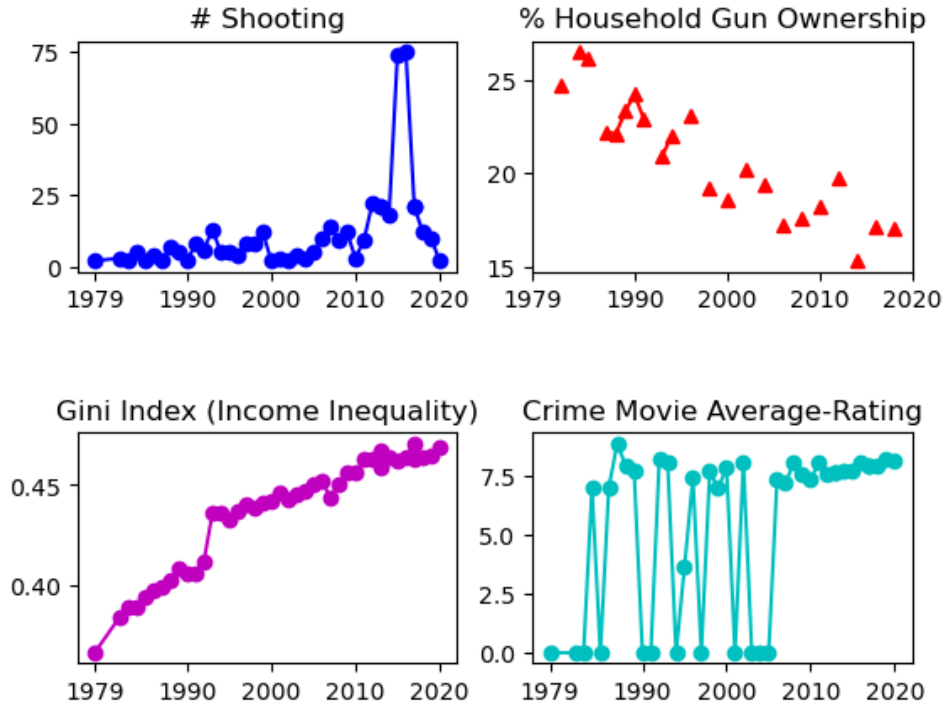


Table 5: Reported Causes that triggered Mass Shooting (**in %**)

Causes	**%**
Mental health problem	21.0
Terrorism	19.0
Anger	13.0
Frustration	6.0
Revenge	3.0
Domestic dispute	4.0
Unemployment	3.0
Others	28.0
Unknown	3.0

[14] https://evergreenpsychotherapycenter.com/violence-media-affects-childrens-behavior/

4. Conclusion

For this project, we created two datasets: Dataset 1- Mass Shooting Cases (1966-2021), and Dataset 2- Aggregated Mass shooting data (1979-2020). Exploratory Data Analysis (EDA) was performed on both datasets to answer some specific research questions.

Our analysis reveals that US mass shootings have an increasing trend, and a significant increase is observed in recent years. Higher shooting rate was observed in February-April months. Most of the shooters are male and majority of them are mid to late teenagers (15-20 years) followed by young and middle adulthood. Half of the shooters are White American or European American (including Hispanic and Latino). Mental health problems should be taken seriously since more than one-third of the shooters were suffering from some mental health issues.

Gun control may reduce gun violence temporarily, but we believe that's not the permanent solution and we need to dig deeper. We explored some selected socio-economic factors and found that income inequality might play a vital role in the issue. Income inequality in USA has increased significantly over the years, which triggers anger and frustration among people. On the other hand, we see from our data that anger, frustration and revenge in the top reported causes of mass shootings. Exposure to excessively violent movies and series which involve attractive and hero-type characters may also shape children's minds to emulate the characters and involve into violent activities like mass shooting.

This project thus recommends experts and policymakers to find and focus on the deep-rooted causes of mass shootings and come up with the ultimate solution to the issue.

5. Appendix A

We created following two datasets using different data sources: Dataset 1- Mass Shooting Cases (1966-2021), and Dataset 2- Aggregated Mass shooting data (1979-2020).

Dataset 1- Mass Shooting Cases (1966-2021):

The Kaggle datasets gave us a great start with lots of data with little work. We combined both datasets by using **pd.concat** method. However, before concatenating, we made sure that both datasets have the same variable name and dropped unnecessary variables from each dataset. Python codes for creating dataset 1 (**dfms1**) are attached below:

```
#importing panda, numpy, matplotlib

import pandas as pd
import matplotlib.pyplot as plt
import numpy as np


#Preparing dataset-1 (massshooting cases): merge two mass shooting data named as "df1" and "df2" from kaggle;
                        #save the dataframe as "dfms1".

    # read in csv file from current folder and load into a pandas Dataframe
df1= pd.read_csv("MassShootings dataset 1.csv")
df2= pd.read_csv("MassShootings dataset 21.csv")

    #have a look at the number of rows, column and number of variables each variables have
df1.columns = df1.columns.str.upper()
df2.columns = df2.columns.str.upper()
#print(len(df1))
#print(len(df2))
#print(df1.head())
#print(df2.head())

    #Create an ID for df2 dataset
#print(df1["S#"])
df2["S#"]=range(324,324+len(df2))
#print(df2["S#"])

    #Change variable names in df2 to match with df1
#print(df1.columns)
#print(df2.columns)
df2=df2.rename(columns={"CASE":"TITLE", "TOTAL_VICTIMS":"TOTAL VICTIMS", "AGE_OF_SHOOTER":"AGE",
                        "PRIOR_SIGNS_MENTAL_HEALTH_ISSUES":"MENTAL HEALTH ISSUES"})
df2["INJURED"]=df2["INJURED"].replace("-","0").astype(int)
df2["TOTAL VICTIMS"]=df2["TOTAL VICTIMS"].replace("-","0").astype(int)
#print(df1.columns)
#print(df2.columns)
#print(df2.head())

    #merge two files
df11=df1[['S#', 'TITLE', 'LOCATION', 'DATE','SUMMARY', 'FATALITIES',
       'INJURED', 'TOTAL VICTIMS', 'AGE','MENTAL HEALTH ISSUES', 'RACE',
       'GENDER', 'LATITUDE', 'LONGITUDE']]
df22=df2[['S#', 'TITLE', 'LOCATION', 'DATE','SUMMARY', 'FATALITIES',
       'INJURED', 'TOTAL VICTIMS', 'AGE','MENTAL HEALTH ISSUES', 'RACE',
       'GENDER', 'LATITUDE', 'LONGITUDE']]
dfms1=pd.concat([df22,df11])
print(len(dfms1))
```

Dataset 2- Aggregated Mass shooting data (1979-2020):

To create the second dataset, we first created an aggregated yearly data from the dataset 1. We first created 'YEAR' variable from 'DATE' to aggregate sum of 'FATALITIES', 'INJURED', and 'TOTAL VICTIMS'. We then combined other socio-economic factors data from external sources to gather more data to analyze and generate important information.

We used CSV data to export and integrate data on 'household gun ownership', 'income inequality', 'average-rating of crime movies and series'. For importing 'poverty rate', we used web scraping. We saved each dataset as a data frame and combined them using **pd.merge** method by year. Python codes for creating dataset 2 (**dfms2y**) are attached below:

```
#Preparing dataset-2 (yearly data): create a yearly dataset from dataet-1 with no. of shooting, fataliies, injured, and vict
                      #merge external datasets (poverty, ) with mass shooting data.

    # create yearly data applying groupby to the 'date' column

    # format dates
dfms1["DATE"] = dfms1["DATE"].astype("datetime64")
dfms1['YEAR'] = dfms1['DATE'].dt.year
dfms1=dfms1[dfms1["YEAR"]!=2022]
#print(len(dfms1))

dfms2=dfms1.groupby("YEAR")["DATE"].count().reset_index(name="# SHOOTING")["# SHOOTING"]


dfms2=pd.DataFrame(dfms2)

dfms2["# FATALITIES"]=dfms1.groupby("YEAR")["FATALITIES"].sum().reset_index(name="# FATALITIES")["# FATALITIES"]
dfms2["# INJURED"]=dfms1.groupby("YEAR")["INJURED"].sum().reset_index(name="# INJURED")["# INJURED"]
dfms2["# VICTIMS"]=dfms1.groupby("YEAR")["TOTAL VICTIMS"].sum().reset_index(name="# TOTAL VICTIMS")["# TOTAL VICTIMS"]

dfms0=dfms1.groupby("YEAR")["DATE"].count().reset_index()
dfms2["YEAR"]=dfms0["YEAR"]

#print(dfms2)

    #Poverty data: yearly (Web scraping)
# Websites to scrape
url1 = 'https://www.self.inc/info/poverty-rates-in-each-state/?fbclid=IwAR3kjQJte-wgZh3K6ay9QAKHlu6Vy8myn7WzSoWc3FVkUcsRrWlD

#Scrape websites and extract HTML tables
dfpoverty = pd.read_html(url1)
#print(dfpoverty )

dfp=dfpoverty[3]
dfp=pd.DataFrame(dfp)
dfp=dfp.rename(columns={"% of people in poverty (total in the U.S.)":"% total poverty"})
dfp=dfp.rename(columns={"Year":"YEAR"})
#print(dfp)


    #Inequality: Gini Index: yearly data (exporting excel file)

        #A higher Gini index indicates greater inequality

dfgi= pd.read_csv("Income Inequality data.csv")
#print(dfgi)

  #Household gun ownership: yearly data (exporting excel file)

dfg= pd.read_csv("Gun ownership.csv")
#print(dfg)
```

```
    #Movie dataset: no. od serial killer/crime movies or series released: maximum IMDB rating by year (exporting excel file)
dfmv= pd.read_csv("Crime-Serialkiller movie liking_1.csv")
dfmv1=dfmv.groupby("YEAR")["RATING"].mean().reset_index()
dfmv1=dfmv1.rename(columns={"RATING":"CRIME MOVIE AVERAGE-RATING"})

    #merge datasets

#print (len(dfms2))
mergedf21=pd.merge(dfms2,dfp, on="YEAR")
#print (len(mergedf21))
mergedf22=pd.merge(mergedf21,dfgi, on="YEAR")
#print (len(mergedf22))
mergedf23=pd.merge(mergedf22,dfmv1, on="YEAR")
#print (len(mergedf23))
dfms2y=pd.merge(mergedf23,dfg, on="YEAR")

dfms2y["% total poverty"]=dfms2y["% total poverty"].str.replace('%', '')
dfms2y.columns=dfms2y.columns.str.upper()
dfms2ycol0 = dfms2y.pop('YEAR')
dfms2y.insert(0, 'YEAR', dfms2ycol0)
```

```
#Final datasets: two datasets
    # 1. Massshooting cases 1966-2021 (each year has multiple cases)

print(dfms1)

    # 2. Aggregated Massshooting data (1984-2020) with other socio-economic indicators

print(dfms2y)

print(dfms2y)
```

<u>We have submitted all the codes on Jupyter Notebook and final datasets with this project report</u>.

6. Appendix B

Contributions of the Group Members:

We all worked as a team for this project and contributed in each part. The most important part we learned from this project is how individual contributions are important in each step to make a good work. For example, one of us was good at generating ideas, one at data manipulation, and other at preparing presentation slides. But, after drafting each of these, other members used to check and verify and improve the work if needed.

Individual Learning from the Project:

1. Group member: Mahmudul Hasan Mamun

As a Statistics person, this project was very interesting for me in many aspects. For data manipulation, I usually use R and SAS. Python was very new to me. For data manipulation and exploratory data analysis for the project, I could use the concepts that I learned in this course. Also, when working with real-world data, we had to deal with different issues that we needed help from external sources to solve.

I understand how important the validity of the data is before anything else. As we know one of the famous quotes- "Garbage in, Garbage out", I wanted to make sure we do not use some garbage data to generate the outputs. In this process, one of the biggest challenges that we faced in the project is to decide on the source of the mass shooting that will be used for the project. Because mass killing has no commonly agreed-upon definition, different sources reported different numbers of mass shootings in the same year. US Congress defines mass shooting as a single shooting incident where three or more people are killed excluding the perpetrator. The gun violence archive, one of the largest and most dedicated sources to record mass shootings, includes all the shootings in which four or more people are shot or killed in a single incident excluding the perpetrator. Though Gun violence archive has a clearly defined methodology for collecting data, the source only for recent years (2014-2023). Since our goal was to analysis trend of historical data, we finally used US mass shooting data from Kaggle, which contains the past 49 years of data until 2021. The author of the dataset consulted several public datasets and web pages to compile this data. Some of the major data sources include Wikipedia, Mother Jones, Stanford, USA Today and other web sources. We did not use 2022 and 2023 data from Gun Violence Archive since they would make anomaly in the dataset and may lead to wrong conclusion.

2. Group member: Jordan Leh

This project was a great learning opportunity for me. While I have a computer science and statistics background from my undergrad, my master's degree will be in Applied Statistics. This class and project combined the two. I am used to data analysis in R and Python, Java, or C for everything else. It was enjoyable to see how applicable Python is to all aspects of my field of study and I look forward to using these skills in my future. This project also gave me real-life experience where data isn't perfect like it is in textbooks.

One interesting element that gave me a new perspective was combining datasets. It was necessary for the datasets to be comparable, such as having the same "definition" of mass shooting. We couldn't just add anything from the Gun Violence archive, or it may make the data inaccurate for certain years when adding the original Kaggle set. Data cleaning is also a new step that I hadn't gotten into much in the past, but I really see the importance now. With textbook or homework data usually, it is ready for analysis but that isn't realistic.

During this project, I worked on a few parts. I helped validate the data and generated ideas for hypotheses. I made sure they were realistic to test based on what we had learned in class. Also, having prior Python experience helped a decent amount, so not everything was unfamiliar. I also helped with the presentation scripts, preparation, and finalizing and formatting the report.

3. Group member: Jordan Wood

My background is in Statistics, Economics, and Psychology. While I have experience in many other programming languages, such as SAS and R, Python was new to me this semester, and it was challenging to begin learning. However, through collaboration with my group, I was able to learn by observing their processes as we worked on the project. As a result, it has changed my perception of Python because it went from being a daunting programming language I felt I would never understand to a helpful tool that can be applied to my other studies.

I thought it was interesting how I was able to take what I learned from this project and apply it to another course. In a graduate-level economics course, I was assigned a research paper where I needed to find a dataset and run an analysis of it. I struggled to find a dataset that met my needs and interests, but I remembered the resources from this class. I was able to find two datasets on Kaggle that had variables I needed and combine them in Python. Then, I was able to use the dataset in my class's software (R Studio) and write my report on a topic I was more interested in since I was able to create the dataset. Overall, I found this project very useful to my understanding of data modeling and am glad to have it as a resource going forward.

7. Appendix C

1. Data source 1: US mass shootings, Kaggle (1966-2017): https://www.kaggle.com/datasets/zusmani/us-mass-shootings-last-50-years
2. Data source 2: US mass shootings, Kaggle (2018-2021): https://www.kaggle.com/datasets/zusmani/us-mass-shootings-last-50-years
3. Data source 3: Gun ownership data: https://www.vpc.org/studies/ownership.pdf
4. Data source 4: Income Inequality data (Gini Index): https://www.census.gov/data/tables/time-series/demo/income-poverty/historical-income-inequality.html
5. Data source 5: Poverty data: https://www.self.inc/info/poverty-rates-in-each-state/?fbclid=IwAR2ViV6vGEN6YSOQA7aUo-_qbKPs5V6mS5aL6gqhcbfSYB9jkYIo3yHeHNg
6. Data source 6: Movie data: https://www.imdb.com/search/title/?title_type=feature,tv_movie,tv_series&user_rating=7.0,10.0&keywords=Serial+Killer+

   https://www.imdb.com/search/title/?title_type=feature,tv_movie,tv_series&user_rating=7.0,10.0&keywords=Crime

7. https://www.census.gov/quickfacts/fact/table/US/IPE120221
8. https://www.bmj.com/company/newsroom/higher-rates-of-mass-shootings-in-us-states-with-more-relaxed-gun-control-laws/
9. https://www.capitalgazette.com/opinion/columns/ac-ce-column-perry-weed-20200823-tvcctkpnrzbihlztoe6q7cffce-story.html
10. https://www.latimes.com/opinion/op-ed/la-oe-sapolsky-air-rage-and-inequality-20160710-snap-story.html
11. https://www.americanprogress.org/article/income-inequality-in-the-united-states-fuels-pessimism-and-threatens-social-cohesion/